



Jennifer C. Cremer

ferbycremer@gmail.com | LinkedIn/jennifer-cremer | <https://ferbycremer.github.io/>

Education

University of Florida, Herbert Wertheim College of Engineering

Ph.D, Computer Graphics & Visualization 2018 – 2026

Relevant Courses: Concurrent Programming, Comp. Graphics, Info. Visualization, Multimodal Data Mining

MSc, Computer Science 2018 – 2021

BS, Digital Arts & Science 2014 – 2018

External focus in Applied Physics: Statics, Thermodynamics, Fluid Mechanics

Research Experience

Scan2Twin (Thesis) – Jörg Peters & Eric Ragan 2018 - 2026

- Refactored orphaned VR software to adhere to proper data handling, object-oriented paradigms, and modern UX guidelines
- Developed C++/OpenGL software for virtual reality (VR) to voxelate medical images and trace out vessels as B-spline curves
- Converted project base from C++/OpenGL/OpenVR to the Unity3D Engine and C#
- Developed VR toolsets for spatial understanding and interactive modeling of voxel data
- Demonstrating functionality and visualization prototypes of anatomy to surgical teams
- Managing project definitions and scope with collaborators in the Colorectal Oncology Team at UF Health: Shands
- Integrated responsive shrink-wrapping methods to transform voxel clouds into discrete meshes
- User study for the impacts of modeling experience on efficiency and accuracy of mesh editing in VR and desktop interfaces
- Investigated capabilities of Skeleton Extraction methods using Pytorch to improve meshing process
- User study exploring the aptitude of various volumetric rendering techniques for different exploratory tasks

Redirected Walking Technique Detection Study – Brett Benda: INDIE Lab, Dept. of CISE, UF 2024

TIPS – Jörg Peters: Surflab 2017 - 2018

- Developed a custom file format to store mesh data as well as scene hierarchy relationships
- Created a Python script for Blender to parse custom files of mesh data and object hierarchy into soft-body surfaces with spring constraints
- Developed user-report interfaces for surgical simulation software with screen captures and descriptions of key training moments using Qt
- Acted as interlocutor for an interdisciplinary team with the UF Veterinary School

Mentoring & Volunteering

SURFLab Undergraduate Student Research Coordinator 2018 - 2026

Project manager to multiple mixed groups of students working on both independent projects and dissertation subprojects

Academy Software Foundation: Diversity & Inclusion Working Group 2020 - Present

- SIGGRAPH Birds of a Feather 2025
- SIGGRAPH Student Volunteer Info Session 2024 – 2026
- Summer Learning Program: Operations Team 2023 – 2025
- DevDays Volunteer 2023
- Summer Learning Program: Team Lead 2021, 2022
- VFX Careers Webinar Series 2021

“VFX Careers: Technical Director” panelist, “University Content: Building from Source with Cmake” presenter

Student Volunteer for ACM SIGGRAPH 2023, 2024

Teaching Experience

Dept. of Computer & Information Science & Engineering, Univ. of FL

Graduate Teaching Assistant

2018 - 2025

Computer Graphics, Information Visualization, Intro to Computational Media, AI for Computer Games, Operating Systems

UF at Kyoto University Summer Abroad Program

2022

Instructor on record

2020 - 2021

Performant Programming in Python (2021), Design Patterns in Object-Oriented Programming (2020, 2021)

Conferences & Papers

Empirical Study of Virtual Reality and Desktop Systems for Qualitative Editing of 3D Meshes: Impacts of Expertise and Context

ACM Symposium on Virtual Reality Software and Technology (VRST 2025)(Conference Paper)

Jennifer C. Cremer, Connor Lausch, Jörg Peters, Eric Ragan

Improving Radiology Communications and Patient Trust with Virtual Reality

ACM Symposium on Virtual Reality Software and Technology (VRST 2025)(Poster)

Jennifer C. Cremer, Connor Lausch, Krista Terracina MD, Jörg Peters, Eric Ragan

Detection of Translation Gain is Decreased when Virtual Reality Users are Unaware of Its Presence

ACM Symposium on Virtual Reality Software and Technology (VRST 2025)(Poster)

Brett Benda, Jennifer C. Cremer, Jogn Fang-Wu, Eric Ragan

Scan2Twin: Virtual Reality for Enhanced Anatomical Investigation

IEEE Conference on Virtual Reality and 3D User Interfaces (IEEEVR 2024) (Doctoral Consortium)

Jennifer C. Cremer

Immersive VR 3D Model for Rectal Cancer Robotic Surgery

American Society of Colon and Rectal Surgeons via Intuitive Research - 2023 - Video Abstract

P. Mazirka, J. Cremer, J. Balch, A. Rashid, K. Ehresmann, L. Goldstein, J. Nordenstam, T.E. Read, J. Grajo, J. Peters, K. Terracina

Patient-Specific MRI VR Model Construction and Simulation

Women in Scientific Computing on Complex Physical and Biological Systems (Poster) - 2022 - Gainesville, FL

Jennifer C. Cremer, Jörg Peters

From Scans & Model Collections to Interactive Surgical Simulation

ACS Surgeons and Engineers 2021 - Poster

Jennifer Cremer, Ruiliang Gao, Krista Terracina MD, Jörg Peters

Languages & Packages

C++, C#, Python, Java, OpenGL, WebGL, HLSL, Bash, JavaScript

Unity 3D, OpenXR, Qt, PyTorch, Adobe CC Suite, Autodesk Maya, Blender

Soft Skills

Mentorship, Interdisciplinary Communication, Project Management, Team Management, Feature Definition

Extracurricular

WarHammer Miniature Painting

2023 – Present

Advanced Open-Water SCUBA Diving – PADI certification

2016 – Present

Amateur Wildlife Photography – birds in flight & macro bugs

2012 – Present